

Dowon Kim

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Education	B.S. in Global School of Media GPA: 4.3/4.5 Soongsil University	Mar 2020 - Feb 2026
Research Experience	LG AI Research Robotics Data Engineer Intern, Physical Intelligence Lab Soongsil University Undergraduate Student Researcher, Reality Lab	Jan 2026 - present Jun 2024 - Dec 2025
Awards	1st Place at ARNOLD Challenge , CVPR2025 Embodied AI Workshop Grand Prize , 2025 Gyeonggi Youth Gap Year Program Outstanding Participant Award , Reading Discussion Seminar Soongsil Univ	Jul 2025 Nov 2025 Aug 2020
Publications	[D2] A Deep Learning-Based Web Application for Automatic Detection and Diagnosis of Skin Diseases , Sebin Lee, Dowon Kim, Geunchan Park, Dohun Kim, Iksu Kim <i>Journal of The Korea Contents Association</i> 2025 [D1] Quantitative Analysis of Multi-Viewpoint Configuration Effects in Robotic Manipulation Policy Learning , Dowon Kim, Sangmin Lee, Sungyong Park, Heewon Kim <i>Journal of Broadcast Engineering</i> 2025	
Projects	Large-Scale Sim-to-Real Robotics Data Pipeline Co-designing and building an end-to-end R2S2R pipeline using NVIDIA Isaac Sim at LG AI Research. Developing the full data loop from real-world collection to synthetic augmentation and deployment validation, focusing on scalable architecture and sim-to-real robustness. Vision-Language-Action Policy Learning on ARNOLD Benchmark Fine-tuned Vision-Language-Action models for embodied manipulation on the ARNOLD benchmark. Improved robustness and generalization through targeted data augmentation, outperforming the PerAct baseline in continuous control tasks within Isaac Sim. Visual Guidance for Failure Modes in VLA Models Analyzing systematic failure modes of VLA models and proposing a visual guidance framework to enhance agent reasoning and control stability across long-horizon manipulation tasks. Humanoid Mobile Manipulation System Design Designed and implemented a humanoid mobile manipulation system to study the full robotics lifecycle, from sim-to-real policy deployment to large-scale dataset generation and evaluation. One-Step Diffusion-based Anomaly Detection Developed a one-step denoising variant of DiffusionAD for real-time structural anomaly detection in biological products, enabling efficient inspection of irregular industrial defects. IoT-based Smart Parking Management System Built a Raspberry Pi-based smart parking system integrating license plate recognition, GPIO-controlled hardware modules, and a Flask-based management interface for real-time monitoring and control.	Jan 2026 - Present Apr 2025 - May 2025 Oct 2025 - Present Aug 2025 - Dec 2025 Mar 2024 - Jun 2024 Sep 2024 - Dec 2024
Scholarship	National Scholarship for STEM (Full Scholarship) , KOSAF	2020 - 2026
Media	Soongsil Univ. 1st Place at CVPR '25 ARNOLD Challenge , The JoongAng	Jul 2025
Skills	Languages: Python, C++, C, Linux Shell Script Tools: Isaac Sim, ROS2, Ubuntu Robot: Franka, RX1, AI worker, OMY	